Term Test, Second Semester 2024

MATHS 350: Topology

(Time allowed: 50 MINUTES)

Each question is worth 10 marks.

1. Let $X = \prod_{\alpha \in A} X_\alpha$ where A is some indexing set, and for each $\alpha \in A$, X_α is a topological space. Show that the Tychonoff topology is the weakest topology on X for which each of the projections $\pi_\alpha : X \to X_\alpha$ is continuous.

2. Let X be a first countable space and let $A \subseteq X$. Prove that

$$\overline{A} = \{x \in X : \text{there is a sequence } (x_n) \text{ in } A \text{ such that } \lim_{x_n \to \infty} = x\}.$$

3. A space X is connected if, and only if, every continuous function $f : X \to \{0, 1\}$ is constant.

4. (a) Suppose that C is a connected subset of X. Prove that $\overline{C}$ is connected.

(b) Is the converse of (a) true? Justify your answer.
